

Raw State PLUG & ILRM

Unnormalised inverse-lag histories, architecture definitions, and
comparative screening experiments

Contact: rickeyernest1990@gmail.com

Publication date: 2 September 2026

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Overview

Raw State PLUG (RS-PLUG) and Raw State ILRM (RS-ILRM) are experimental variants of PLUG and ILRM in which the inverse-lag history is used **without harmonic normalisation**. The change is deliberately small. The lag kernel remains exactly

$$1, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{1}{4}, \quad \dots$$

and the history is still constructed from **previous input representations only**. The only defining change is that the accumulated history is not divided by the harmonic number.

This document uses “Raw State” to name that unnormalised deterministic history. It does **not** mean that the recurrent hidden state itself is exposed, unbounded, or otherwise changed.

The most important mathematical fact is also the most sobering one: at a fixed timestep, the raw and normalised histories differ only by a positive scalar. If the normalised history

is P_t and the raw history is R_t , then

$$R_t = H_t P_t.$$

Therefore the raw construction does not create extra information or unmix already-superposed inputs. Its potential benefit comes from changing the scale presented to nonlinear recurrent gates and from removing the additional H_t attenuation from the direct input-to-history gradient path.

Experimental status. The results in this document are controlled synthetic screening experiments. They are not peer-reviewed evidence of general superiority over GRU, the original PLUG/ILRM forms, Transformers, state-space models, or other architectures. The strongest result is a task-dependent persistence benefit; the clearest limitation is multi-item interference and poor exact sequence reconstruction.

1 Motivation and scope

The original PLUG and ILRM definitions use a normalised inverse-lag history so that the coefficients sum to one. This keeps the history in the scale of a weighted average and is well behaved even when input representations have a persistent non-zero mean.

During follow-up screening, removing the harmonic denominator sometimes improved long-range accuracy and training reliability. Other tests showed the opposite: the normalised history could be more stable when several relevant items had to coexist. This document formalises that ablation and reports both positive and negative results.

The intended research question is narrow:

What changes when the same $1/k$ history is supplied to PLUG or ILRM at its raw accumulated scale rather than as a harmonic weighted average?

2 Normalised inverse-lag history: baseline

Let the current input representation be $x_t \in \mathbb{R}^{d_x}$. Define the harmonic number

$$H_t = \sum_{k=1}^t \frac{1}{k}.$$

The original PLUG/ILRM history is

$$P_t = \frac{1}{H_t} \sum_{k=1}^t \frac{x_{t-k}}{k}, \quad P_0 = 0.$$

The current input x_t is excluded. For an input k positions in the past,

$$\frac{\partial P_t}{\partial x_{t-k}} = \frac{1}{k H_t} I.$$

For the first input x_0 , the direct history coefficient is $1/(tH_t)$, which is approximately $1/(t \log t)$ at large t .

3 Raw State inverse-lag history

Raw State removes only the scalar normaliser:

$$R_t = \sum_{k=1}^t \frac{x_{t-k}}{k}, \quad R_0 = 0.$$

The first few states are

$$\begin{aligned} R_1 &= x_0, \\ R_2 &= x_1 + \frac{1}{2}x_0, \\ R_3 &= x_2 + \frac{1}{2}x_1 + \frac{1}{3}x_0. \end{aligned}$$

The kernel is still exactly $K(k) = 1/k$. No learned exponent, gate, attention mechanism, extra memory bank, or additional trainable state is introduced.

4 Relationship between raw and normalised states

Because the normaliser is a scalar,

$$R_t = H_t P_t.$$

This has several consequences.

- At a fixed t , R_t and P_t point in exactly the same direction in feature space.
- Raw State does not improve linear source separation by itself. The same old inputs remain superposed with the same relative $1/k$ weights.
- Any performance difference must arise from scale-sensitive downstream computation, optimisation, nonlinear saturation, gating, or sequence-length extrapolation.
- A scale-invariant downstream model with exact knowledge of t could, in principle, convert one form into the other by multiplying or dividing by H_t .

This distinction is important when interpreting the experiments: Raw State is a **scaling ablation**, not a new information channel.

5 Scale, variance, and non-zero-mean risk

Suppose one input coordinate is approximately zero mean, independent across time, and has variance σ^2 . Then

$$\text{Var}(R_t) = \sigma^2 \sum_{k=1}^t \frac{1}{k^2}.$$

Since

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6},$$

the raw history has bounded variance under this idealised zero-mean model. By contrast,

$$\text{Var}(P_t) = \frac{\text{Var}(R_t)}{H_t^2},$$

so the normalised history’s RMS scale decreases roughly as $1/\log t$.

This does **not** mean Raw State is always better behaved. If inputs have persistent mean μ , then

$$\mathbb{E}[R_t] = \mu H_t \approx \mu(\log t + \gamma),$$

whereas $\mathbb{E}[P_t] \approx \mu$. Raw State can therefore develop a slowly growing common-mode component on non-zero-mean or strongly correlated streams. That risk is one plausible reason the normalised versions perform better on some interference-heavy tests.

t	H_t	raw oldest coefficient $1/t$	normalised oldest coefficient $1/(tH_t)$
64	4.744	1.562e-02	3.294e-03
512	6.817	1.953e-03	2.865e-04
1,024	7.509	9.766e-04	1.300e-04
4,096	8.895	2.441e-04	2.745e-05
8,192	9.588	1.221e-04	1.273e-05

At $t = 4096$, Raw State presents the oldest direct history contribution at about 8.90 times the normalised amplitude. At $t = 8192$, the factor is about 9.59.

6 Direct gradient path

For Raw State,

$$\frac{\partial R_t}{\partial x_{t-k}} = \frac{1}{k}I,$$

and for the first input,

$$\frac{\partial R_t}{\partial x_0} = \frac{1}{t}I.$$

This removes the additional logarithmic attenuation introduced by harmonic normalisation. The direct path therefore scales as $1/t$ for the oldest input instead of approximately $1/(t \log t)$.

This is only a direct history-path statement. It does **not** guarantee a non-vanishing end-to-end loss gradient, stable recurrent Jacobians, unlimited memory, or successful decoding after many signals have been mixed into a finite-dimensional state.

7 Raw State PLUG

PLUG changes the input supplied to an otherwise ordinary GRU. Raw State PLUG forms

$$e_t = [x_t; R_t]$$

and feeds e_t to a conventional GRU:

$$\begin{aligned} r_t &= \sigma(W_{ir}e_t + b_{ir} + W_{hr}h_{t-1} + b_{hr}), \\ z_t &= \sigma(W_{iz}e_t + b_{iz} + W_{hz}h_{t-1} + b_{hz}), \\ n_t &= \tanh(W_{in}e_t + b_{in} + r_t \odot (W_{hn}h_{t-1} + b_{hn})), \end{aligned}$$

$$h_t = (1 - z_t) \odot n_t + z_t \odot h_{t-1}.$$

The GRU recurrence itself is unchanged. The defining difference from original PLUG is simply R_t in place of P_t .

8 Raw State ILRM

Raw State ILRM keeps the same two-gate ILRM cell and replaces only the history input. Form

$$z_t = [x_t; h_{t-1}; R_t].$$

Then compute

$$w_t = \sigma(W_w z_t + b_w), \quad r_t = \sigma(W_r z_t + b_r),$$

construct

$$\tilde{h}_t = \tanh(W_c[x_t; r_t \odot h_{t-1}; R_t] + b_c),$$

and update

$$h_t = (1 - w_t) \odot h_{t-1} + w_t \odot \tilde{h}_t.$$

There is still no third gate, no LSTM-style cell state, no attention, no external key-value memory, and no learned decay exponent.

9 Parameter dimensions and implementation cost

Raw State adds no trainable parameters relative to the corresponding normalised architecture. It removes one scalar division per history timestep after the convolution has been computed.

In the main $d_x = 16$ binary screening suite, the closely matched models contained:

Model	Trainable parameters	Relative to GRU
GRU	1,649	baseline
PLUG / RS-PLUG	1,669	+1.2%
ILRM / RS-ILRM	1,633	-1.0%

The output head changes the exact total in multi-class tasks, but normalised and Raw State versions remain parameter-identical within each architecture.

10 Efficient history calculation with FFT

Define the fixed causal kernel

$$q = [0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots].$$

Then Raw State is simply the causal linear convolution

$$R = X * q.$$

For a sequence of length T , choose a padded FFT length $L \geq 2T - 1$, compute the linear convolution by zero-padded FFT, and keep the first T causal samples. The leading zero remains critical because it prevents the current input from leaking into R_t .

The normalised implementation performs one extra operation:

$$P_t = R_t / H_t.$$

Raw State stops before that division.

11 Minimal PyTorch-style reference implementation

```
import torch
import torch.nn as nn

def inverse_lag_history_raw_reference(x):
    # x: [batch, time, input_size]
    # R[:, t] contains inputs strictly before timestep t.
    B, T, D = x.shape
    R = torch.zeros_like(x)
    for t in range(1, T):
        numerator = torch.zeros_like(x[:, 0])
        for k in range(1, t + 1):
            numerator = numerator + x[:, t-k] / k
        R[:, t] = numerator
    return R

class RawStatePLUG(nn.Module):
    def __init__(self, input_size, hidden_size):
        super().__init__()
        self.gru = nn.GRU(2 * input_size, hidden_size,
                           batch_first=True)

    def forward(self, x, h0=None):
        R = inverse_lag_history_raw_reference(x)
        e = torch.cat([x, R], dim=-1)
        return self.gru(e, h0)
```

For RS-ILRM, use the same R_t in the original ILRM gate and candidate equations shown above. A production implementation should use the verified zero-padded FFT history rather than the quadratic reference loop.

12 Experimental protocol

The new comparison screens were designed to compare five models under paired conditions:

- standard GRU;
- original PLUG with $P_t = R_t / H_t$;
- RS-PLUG with R_t ;

- original ILRM with $P_t = R_t/H_t$;
- RS-ILRM with R_t .

Unless stated otherwise, the principal distance screens used 10 paired seeds, Adam with learning rate 0.003, batch size 64, 250 optimiser updates, and training lengths sampled only from 32–64. Evaluation then extrapolated without retraining to 64, 128, 256, 512, 1,024, 2,048, 4,096, and 8,192 steps.

Evaluation sample count was reduced at the longest endpoints to control CPU cost: 512 examples per seed through 512, 256 at 1,024, 128 at 2,048, 64 at 4,096, and 32 at 8,192. The 8,192-point results should therefore be treated as exploratory.

Inputs were structured numeric feature channels, not tokenised natural language and not real market data. These experiments test recurrent persistence, interference, binding, and optimisation behaviour in controlled synthetic settings. They do not establish application-level accuracy.

13 Comparison experiments

13.1 Direct normalisation ablation: fact recall

A first 10-seed paired screen compared only the four history architectures while holding the architecture and training generator fixed. Values are mean accuracy $\pm$ one standard deviation.

Model	64	128	256	512
PLUG	69.7 $\pm$ 23.5	67.8 $\pm$ 23.1	66.6 $\pm$ 23.6	66.5 $\pm$ 23.3
RS-PLUG	86.2 $\pm$ 21.9	85.7 $\pm$ 22.7	84.9 $\pm$ 24.0	84.8 $\pm$ 24.2
ILRM	99.5 $\pm$ 1.4	99.5 $\pm$ 1.5	99.5 $\pm$ 1.4	99.5 $\pm$ 1.5
RS-ILRM	100.0 $\pm$ 0.1	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0

At 512 steps, RS-PLUG averaged 84.8% versus 66.5% for normalised PLUG. RS-ILRM was at ceiling (100.0%) while normalised ILRM averaged 99.5%. The large PLUG standard deviations indicate a training-stability effect: some seeds learned the task essentially perfectly while others failed.

13.2 Direct normalisation ablation: fact correction

The same four-way ablation was repeated on a correction task in which a stored fact had to be replaced later.

Model	64	128	256	512
PLUG	100.0 $\pm$ 0.0	93.6 $\pm$ 20.3	90.7 $\pm$ 20.9	78.4 $\pm$ 23.8
RS-PLUG	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	99.9 $\pm$ 0.2	92.4 $\pm$ 15.1
ILRM	100.0 $\pm$ 0.0	94.2 $\pm$ 11.2	85.9 $\pm$ 22.8	86.2 $\pm$ 22.3
RS-ILRM	100.0 $\pm$ 0.0	95.9 $\pm$ 12.8	94.9 $\pm$ 16.2	93.4 $\pm$ 17.6

At 512, RS-PLUG improved from 78.4% to 92.4% and RS-ILRM from 86.2% to 93.4% on average. Seed-to-seed variability remained substantial, so these are screening results rather than a universal claim.

13.3 Long-distance single-fact recall

This test stored one task-relevant binary fact among structured distractors. Chance was 50%. The models were trained only on lengths 32–64.

Model	64	512	1,024	4,096	8,192
GRU	48.7 ± 1.9	49.4 ± 2.6	49.2 ± 2.8	54.5 ± 4.2	52.8 ± 10.3
PLUG	93.8 ± 15.0	90.5 ± 20.0	90.3 ± 20.4	91.4 ± 18.2	88.8 ± 23.8
RS-PLUG	93.1 ± 14.8	90.3 ± 20.3	90.1 ± 20.9	90.8 ± 19.6	89.7 ± 21.8
ILRM	99.9 ± 0.3	99.9 ± 0.2	99.8 ± 0.6	100.0 ± 0.0	100.0 ± 0.0
RS-ILRM	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0

GRU remained near chance. Both ILRM forms were essentially perfect through the tested range. PLUG and RS-PLUG were both around 90% at long distance; in this generator Raw State did **not** materially outperform normalisation. This is a useful counterexample to the earlier direct ablation.

13.4 Oldest-item associative binding

Four entities received independent values in random order. The query was restricted to one of the first three presented entities so that remembering only the most recent item could not produce a shortcut. Chance was 50%.

Model	64	512	1,024	4,096	8,192
GRU	50.4 ± 2.9	50.8 ± 1.7	49.7 ± 4.0	50.9 ± 4.2	48.8 ± 10.8
PLUG	70.5 ± 2.1	63.7 ± 6.5	63.6 ± 7.6	61.4 ± 11.6	57.8 ± 9.9
RS-PLUG	71.9 ± 2.0	65.7 ± 2.9	61.5 ± 5.8	60.0 ± 6.1	56.6 ± 9.4
ILRM	71.3 ± 2.2	64.9 ± 6.6	62.7 ± 8.9	60.9 ± 7.1	59.1 ± 11.3
RS-ILRM	71.8 ± 1.9	63.7 ± 8.4	58.0 ± 8.6	56.2 ± 8.0	50.9 ± 10.3

All history models beat GRU at short and medium range, but binding degraded with distance. At 8,192, normalised ILRM averaged 59.1% versus 50.9% for RS-ILRM; normalised PLUG averaged 57.8% versus 56.6% for RS-PLUG. The result suggests that stronger history amplitude does not solve relational collision and may worsen it in ILRM.

13.5 Mutable state / fact update

An entity received an early value and later an independently sampled replacement value. The correction location varied across the sequence. The final query asked for the current value.

Model	64	512	1,024	4,096	8,192
GRU	100.0 ± 0.0	90.4 ± 20.3	89.8 ± 21.4	90.8 ± 19.5	91.9 ± 17.2
PLUG	100.0 ± 0.0	95.0 ± 15.9	92.2 ± 16.8	85.6 ± 23.3	83.4 ± 26.8
RS-PLUG	100.0 ± 0.0	85.5 ± 23.4	84.4 ± 25.2	86.6 ± 21.7	85.0 ± 24.2
ILRM	100.0 ± 0.0	97.0 ± 7.5	96.9 ± 7.5	95.9 ± 8.8	96.6 ± 8.9
RS-ILRM	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0

RS-ILRM reached 100% mean accuracy at every tested distance through 8,192 in this screen. Normalised ILRM remained very strong at 96.6% at 8,192. PLUG behaved differently: normalised PLUG was better at 512 and 1,024, while the two PLUG forms converged to the mid-80s by the longest endpoints. This is strong task-specific evidence for RS-ILRM, not evidence that Raw State is universally superior.

13.6 Five-event majority aggregation

Five marked binary events were distributed through the stream and the model had to report their majority at the end. This probes accumulation of distributed evidence rather than exact recall of one token.

Model	64	512	1,024	4,096	8,192
GRU	98.6 $\pm$ 3.2	74.2 $\pm$ 5.0	68.4 $\pm$ 6.4	60.6 $\pm$ 9.0	60.3 $\pm$ 10.9
PLUG	99.7 $\pm$ 0.3	81.4 $\pm$ 6.5	70.9 $\pm$ 8.9	58.0 $\pm$ 10.6	54.7 $\pm$ 12.7
RS-PLUG	99.7 $\pm$ 0.3	77.7 $\pm$ 9.2	69.4 $\pm$ 8.2	62.5 $\pm$ 11.8	60.9 $\pm$ 9.2
ILRM	98.9 $\pm$ 1.1	69.8 $\pm$ 8.6	57.5 $\pm$ 5.3	53.3 $\pm$ 5.7	49.4 $\pm$ 7.6
RS-ILRM	99.2 $\pm$ 0.4	76.6 $\pm$ 7.2	68.0 $\pm$ 8.6	61.1 $\pm$ 11.8	62.2 $\pm$ 10.4

Raw State helped ILRM substantially at long distance: at 1,024, RS-ILRM averaged 68.0% versus 57.5% for normalised ILRM; at 8,192 the comparison was 62.2% versus 49.4%. For PLUG the picture was mixed: normalised PLUG was stronger at 512, while RS-PLUG was stronger at 4,096 and 8,192. This is consistent with a persistence-versus-scale-control trade-off rather than a uniformly better formulation.

13.7 Selective two-item memory

Two independent relevant bits were stored: an early item A and a later item B. The final query asked for either one. Overall accuracy can hide a one-slot strategy, so Query A and Query B were also measured separately.

Model	Overall 64	Overall 512	Overall 1,024	Overall 4,096
GRU	66.5	65.0	63.7	63.1
PLUG	74.6	74.9	69.1	69.7
RS-PLUG	75.7	63.8	58.8	49.4
ILRM	75.2	67.9	67.0	67.5
RS-ILRM	75.1	72.6	71.7	58.8

At 4,096, the slot-specific means were:

Model	Query A	Query B
GRU	74.9	54.2
PLUG	73.8	64.2
RS-PLUG	61.0	38.2
ILRM	77.7	58.4
RS-ILRM	70.5	50.8

Normalised PLUG produced the most balanced 4,096-step result (73.8% on A and 64.2% on B). RS-PLUG retained A better than B (61.0% versus 38.2%), and RS-ILRM also favoured A (70.5% versus 50.8%). This is direct evidence that stronger persistence can come at the cost of accepting or separating later relevant information.

13.8 Exact three-symbol sequence reconstruction

Three random symbols had to be reconstructed in order after a long distractor stream. Token-level chance was 12.5%; exact three-symbol sequence chance was about 0.195%.

Model	Token 64	Token 512	Token 1,024	Token 4,096
GRU	12.3	12.6	13.3	11.6
PLUG	12.4	12.7	12.9	12.5
RS-PLUG	17.9	16.7	16.3	17.0
ILRM	12.5	12.6	12.4	12.3
RS-ILRM	14.5	14.3	13.6	14.8

The normalised models and GRU stayed near token chance. Raw State produced modestly more recoverable token information, especially RS-PLUG (17.0% token accuracy at 4,096), but exact-sequence accuracy remained essentially zero. This is a clear negative result: these compressed recurrent histories are not substitutes for exact token-addressable memory.

13.9 Same-channel event-memory stress screen

An early signed event was followed by continuous random values and later distractor events occupying the same semantic event channel. The final query asked for the original event. This screen was deliberately hostile to simple persistence.

Model	64	512	1,024	4,096	seeds >90% at 4,096
GRU	49.6	49.2	50.1	50.0	0/10
PLUG	51.2	49.3	49.3	48.4	0/10
RS-PLUG	59.5	54.8	54.8	53.8	1/10
ILRM	62.1	59.8	60.5	61.9	2/10
RS-ILRM	58.0	54.8	55.4	56.2	1/10

The averages were modest, but the seed distribution was bimodal: some history-model seeds discovered a near-perfect solution that persisted through 4,096, while many remained near chance. Normalised ILRM had two such seeds, RS-ILRM one, and RS-PLUG one. The most defensible interpretation is not a reliable event-memory advantage but an **optimisation reliability problem**: the architecture can represent a solution in some runs, yet the small training budget does not reliably find it.

14 What the experiments currently support

The combined screening evidence supports the following narrow claims:

- Removing harmonic normalisation changes the direct history coefficient from $1/(kH_t)$ to $1/k$ and removes a sequence-length-dependent shrinkage factor seen by the recurrent network.
- Raw State can materially improve performance on some persistence-heavy tasks. The clearest examples in these screens are PLUG fact-recall/fact-correction ablations, ILRM five-event aggregation at long distance, and the mutable fact-update screen for RS-ILRM.
- Raw State does not add parameters and does not change the relative $1/k$ lag weighting.
- Both normalised and raw history architectures can extrapolate far beyond their 32–64 training lengths on some controlled tasks.
- The normalised forms can be better when several relevant items must coexist or when later information must remain separable from earlier information.

15 What the experiments do not support

The experiments do **not** support the following stronger claims:

- They do not show that Raw State is universally better than normalised PLUG or ILRM.
- They do not show that Raw State contains more linear information at a fixed timestep; $R_t = H_t P_t$ exactly.
- They do not solve general source separation, associative binding, or multi-item interference.
- They do not provide exact token memory. The three-symbol copy screen remained close to chance at the token level for most variants and essentially zero in exact sequence recall.
- They do not establish natural-language chatbot quality, real financial-market prediction, sensor reliability, or production robustness. The inputs here are structured synthetic channels.
- They do not demonstrate unlimited memory. The finite-dimensional history still superposes many inputs, and optimisation, interference, scale drift, and capacity can all become limiting factors.
- The 8,192-step endpoints use fewer evaluation examples and should be treated as exploratory.

16 Interpretation: a persistence–interference trade-off

The most coherent interpretation of the current results is a scale trade-off.

Normalisation makes the history a bounded weighted average for bounded inputs and protects against common-mode growth. It also shrinks the entire history approximately as $1/H_t$ relative to Raw State as sequence length increases.

Raw State keeps the full accumulated amplitude. This can make an old signal easier for nonlinear gates to detect at long distance and strengthens the direct gradient coefficient by a factor of H_t . However, because the superposition pattern itself is unchanged, every other component is strengthened by the same scalar. When multiple relevant items compete inside the same finite-dimensional history, the larger amplitude can make nonlinear interference or saturation worse rather than better.

A useful summary is therefore:

Raw State: stronger persistence and scale $\leftrightarrow$ Normalised: stronger scale control
--

Which side is preferable appears to depend on whether the task is dominated by retaining one persistent signal or by separating multiple simultaneously useful signals.

17 Reproducibility checklist

An independent implementation of the Raw State variants should preserve the following points:

1. R_t is computed from previous input representations, never previous hidden states.
2. The lag kernel is exactly $1/k$.
3. There is **no** division by H_t in the Raw State variants.
4. $R_0 = 0$ and the current input x_t is excluded from R_t .
5. RS-PLUG forms $e_t = [x_t; R_t]$ and uses an otherwise unchanged standard GRU.
6. RS-ILRM replaces P_t with R_t in both gate inputs and candidate construction; all ILRM recurrent equations are otherwise unchanged.
7. Normalised and Raw State models must use identical hidden widths when testing the effect of normalisation alone.
8. Parameter matching against GRU should adjust hidden width rather than giving one architecture a large capacity advantage.
9. FFT implementations must use a leading-zero kernel and enough zero-padding to compute linear, not circular, convolution.
10. New FFT code should be checked against a direct float64 reference calculation before benchmarking.
11. Paired comparisons should use the same generated training/evaluation examples and the same random seeds where possible.
12. Long-distance endpoints with reduced evaluation counts should be labelled exploratory.

18 Short implementation-ready definitions

Raw State PLUG

At timestep t ,

$$R_t = \sum_{k=1}^t \frac{x_{t-k}}{k}, \quad R_0 = 0.$$

Form

$$e_t = [x_t; R_t]$$

and feed e_t to an otherwise unchanged standard GRU.

Single most important rule: Raw State PLUG changes the history scale supplied to the GRU; it does not change the GRU recurrence.

Raw State ILRM

At timestep t ,

$$R_t = \sum_{k=1}^t \frac{x_{t-k}}{k}, \quad R_0 = 0.$$

Form

$$z_t = [x_t; h_{t-1}; R_t],$$

compute

$$w_t = \sigma(W_w z_t + b_w), \quad r_t = \sigma(W_r z_t + b_r),$$

construct

$$\tilde{h}_t = \tanh(W_c[x_t; r_t \odot h_{t-1}; R_t] + b_c),$$

and update

$$h_t = (1 - w_t) \odot h_{t-1} + w_t \odot \tilde{h}_t.$$

Single most important rule: Raw State ILRM keeps the original two-gate ILRM recurrence and changes only the history from R_t/H_t to R_t .

19 Publication, contact and licensing

Publication date: 2 September 2026

Contact: rickeyernest1990@gmail.com

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